

If the file *mfc100u.dll* is missing, it's impossible to run *imdisplay*. Simply download and install Microsoft Visual C++ 2010 SP1 Redistributable Package. Some useful common operations are summarised in the following code from the command prompt. ImageMagick is very powerful and has many capabilities. More examples can be found at <http://www.imagemagick.org/Usage>

```
- Reduce image quality/size
  convert -strip -quality 25 big_image.jpg new_image.jpg
- Generate a canvas with a specific size and color
  convert -size 1280x800 canvas:"#708090" background.jpg
- Convert from PNG to ICO
  convert -define icon:auto-resize=256,128,64,48,32,16 file.
  png file.ico
- Convert from PDF or EPS to PNG
  convert -density 300 -quality 100 -layers flatten file.pdf
  file.png
- Convert from color to gray scale
  convert -colorspace Gray color_image.png gray_image.png
- Modulate Brightness, Saturation, Hue
  convert -modulate 100,100,-50 image.png new_image.png
- Resize
  mogrify -resize 128x256 image.png
```

Beside the final conclusions, I would like to add some remarks about these three kinds of software: database, CAD and RAD. For a database on a network, I prefer to use MySQL/MariaDB with some PHP (see the *OSFY* August 2015 issue) and for Drupal websites too. But for a small, desktop only application the story is different. I like well-known commercial software, but find them expensive. But one well-known commercial software that is not too costly is LibreOffice Base. It is an interesting but, in my opinion, not so good alternative to the above-mentioned commercial software, because it still needs a lot of work before it can

be considered a true alternative. The Java engine is slow; Firebird has some stability problems and the GUI is not the best. So, actually for me, this is an open topic.

About CAD, I have no experience about 3D so I would like to list six interesting free alternatives to AutoCAD LT and to VectorWorks (I haven't used them both for years): DoubleCAD XT, DraftSight, FreeCAD, Medusa, nanoCAD and Solid Edge 2D. Some of these options also have 3D capabilities but only FreeCAD is open source. I've tested only Solid Edge (because it's used in the CAD/CAE course at my college) and DraftSight, and I think that the second, for my simple needs, can be an alternative because it's available for GNU Linux (note that DraftSight is unsupported on Linux 32-bit machines).

Last, some short notes about NetBeans and Lazarus. As well as in the case of text editors, I can't make a definitive choice because Java is very widely used and having some knowledge about it has certain advantages, but I also like the Pascal syntax. At this point a big section could be open to talk about the Python programming language on Windows. But this is another story. Finally, can Windows be a good base for free and open source software? I think it can, because on Windows, I can install many other free software (even if not open source) or, if necessary, commercial software that's not available for GNU Linux. But for the future, I hope that it will be possible to use only GNU Linux. 

References

- [1] <http://mbmccllelland.blogspot.it/2012/11/creating-free-pdf-printer-in-windows-7.html>, last visited on 28/08/2015.
- [2] <http://www.stat.tamu.edu/~henrik/GSWriter/GSWriter.html>, last visited on 28/08/2015.

By: Stefano Rizzi

The author works in the analytical chemistry and textile chemistry areas. He has been a Linux user since 1998.

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